

Over vector can never be an eigen vector
therefore we try to find the
non trivial sol.

day / date: 15/04/2025

LINEAR ALGEBRA - ES205.

Chapter no. 5

Ex: 5.1 :-

The stage matrix model
is a diff equation of the
form $x_{k+1} = Ax_k$ — (1)

where x_{k+1} & x_k are the
versions of the same variable
Such an equation is called
a dynamical system.

(because it describes
the change in a system
as the time passes).

DEFINITION :-

An eigen vector of an
 $n \times n$ matrix A is a non
zero vector x such that

$$Ax = \lambda x$$

for some scalar λ .

A scalar λ is called
eigen value of A if there
is a nontrivial solution
 x of $Ax = \lambda x$; such an
 x is called an eigen-
vector corresponding to
 λ .

Example: - 1

$$\text{Let } A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$$

$$u = \begin{bmatrix} 6 \\ -5 \end{bmatrix} \quad \& \quad v = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$\therefore$ it is easy to determine
if u or v are the eigen
vectors of A by using the
definition.

$$Au = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ -5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 6 - 30 \\ 30 - 10 \end{bmatrix} \Rightarrow \begin{bmatrix} -24 \\ 20 \end{bmatrix}$$

$$\Rightarrow -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix}$$

$$\Rightarrow -4u$$

it satisfies the form

$$Ax = \lambda x ; \therefore \text{Yes!}$$

it is an eigen
vector of A .



NOTE:- $Ax = \lambda x$ $\rightarrow$ it causes the direction of the vector x to remain same rather only scaling it.

Similarly:-

$$Av = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$\Rightarrow \begin{matrix} 3 - 12 \\ 15 - 4 \end{matrix} \Rightarrow \begin{matrix} -9 \\ 11 \end{matrix}$$

$Av \neq \lambda v$
 $\therefore$ it is not an eigenvector!

EXAMPLE

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$$

$$Ax = 7x$$

determine the eigenvectors $\Rightarrow$

$$Ax - 7x = 0$$

$$(A - 7I)x = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix} x = 0$$

solving matrix equ!

$$\begin{bmatrix} -6 & 6 & 0 \\ 5 & -5 & 0 \end{bmatrix}$$

$$\sim R_1 \rightarrow R_1$$

$$\sim R_2 \rightarrow R_2$$

$$\begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix}$$

$$R_2 - R_1 \rightarrow R_2$$

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ general solution}$$

where $x_2 \neq 0$.

Note:

This method of echelon form of matrix A usually does not display edge in values of A ; only used for eigenvectors.



Class note: - $Ax = \lambda x$; How many linearly independent eigenvectors in the sol of (1) mean ~~any~~ ^{ant} dimensional space (eigen space is collection of all the eigen vectors)

Note:

Example

if (a) is eigen vect
then all other points generated are eigen space.

$\lambda =$ is an eigenvalue of an $n \times n$ matrix A iff the equation.

$$A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}; \text{eigenvalue} \Rightarrow 2.$$

$(A - \lambda I)x = 0 \rightarrow$ (a) has a nontrivial solution

find Basis of the corresponding eigen space.

The set of all the solutions of (a) is just the Null space of the matrix $(A - \lambda I)$, $\therefore$ this set is a subspace of $\mathbb{R}^n$ and called the eigen space of A corresponding to λ

$$A - \lambda I \Rightarrow A - 2I = 0.$$

$$\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 8 \\ 2 & -1 & 8 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

Eigen space consists of zero vector and all the vectors corresponding to λ .

Solving matrix

$$\begin{bmatrix} 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \end{bmatrix}$$

In the prev e.g; the eigen space is all the multiples of $(1, 1)$ which is the line $(1, 1)$ and the origin.

$$\sim \begin{bmatrix} 2 & -1 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

where x_2 & x_3 are free variables.

The general solution:-

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1/2 \\ 0 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

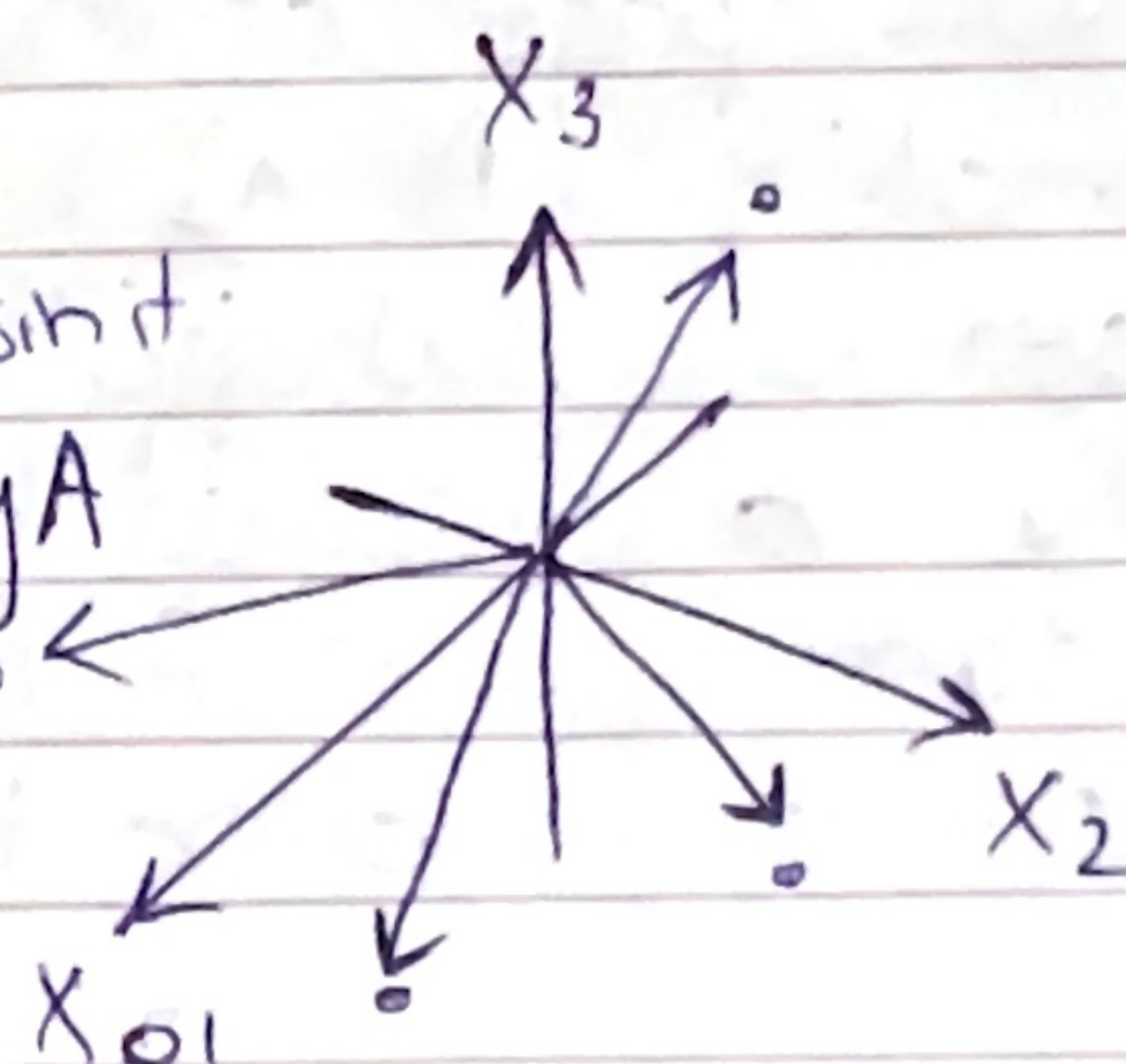
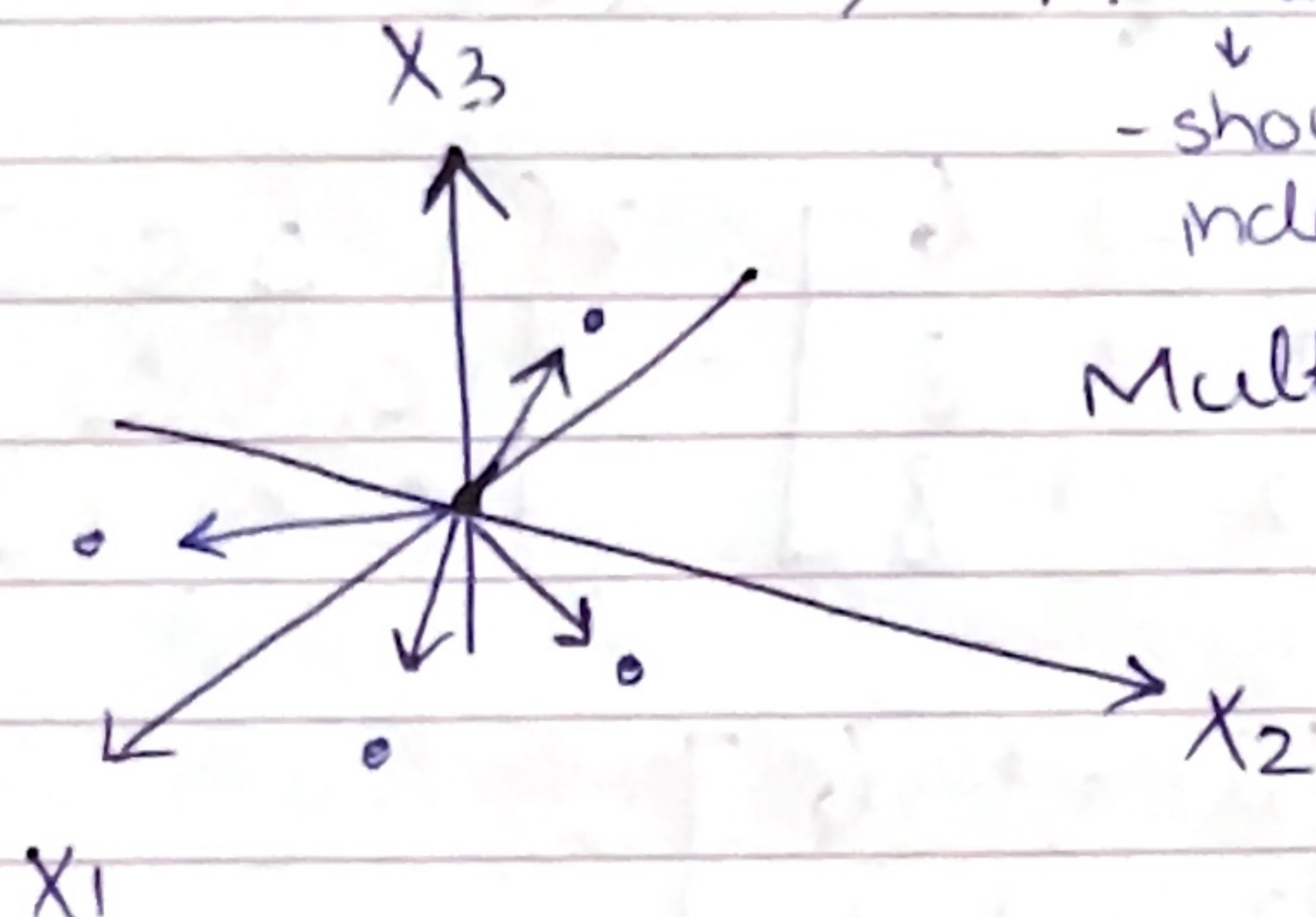
$$\Rightarrow x_2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a 2 dimensional subspace of $\mathbb{R}^3$!

$\hookrightarrow$ A basis!

$\downarrow$
- should be linearly independent vectors in it.

Multiplication by A



What are the eigen vector of I_n ?

$$Ax = \lambda x$$

$$\textcircled{1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 1 \begin{bmatrix} x \\ y \end{bmatrix}$$

where $x, y \in \mathbb{R}$

also knowing the scaling matrix

$$\textcircled{2} \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = k \begin{bmatrix} x \\ y \end{bmatrix}$$

where $x \& y \in \mathbb{R}$.

Note a special case of

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \textcircled{3}$$

if $Ax = \lambda x$ for

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \text{ then.}$$

$$Ax \Rightarrow \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ 2y \end{bmatrix} \neq \lambda x \quad \therefore \begin{bmatrix} x \\ y \end{bmatrix} \text{ cannot be eigen vectors}$$

But if the eigenvectors were to be either

$$\begin{bmatrix} 0 \\ y \end{bmatrix} \text{ or } \begin{bmatrix} x \\ 0 \end{bmatrix} \text{ they}$$

would satisfy from $Ax = \lambda x$

④ Now note the case of flipping matrix of chapter 4.

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Does all the points of only y-axis for an eigen vector exist (i.e. is $\begin{bmatrix} 0 \\ y \end{bmatrix}$ the eigen vector of A) $\Rightarrow$ Yes!

similarly.

Does all the points of only x-axis for an eigen vector exist (i.e. is $\begin{bmatrix} x \\ 0 \end{bmatrix}$ the eigen vector of A)?

Yes!

$\rightarrow$ you can check both by solving as well.

THEOREM NO. 1:-

The eigen values of a triangular matrix are the entries on the main diagonal.

Example.

$$\textcircled{1} \begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix} = A$$

The eigen values of A are 3, 0, 2.

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow \lambda_1 = 1, \lambda_2 = 2, \quad \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \Rightarrow \lambda_1 = 1, \lambda_2 = 3, \quad \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$$

$$(2) \begin{bmatrix} 4 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & 3 & 4 \end{bmatrix} = B$$

Eigenvalues of B are
4, 1, 4.

What does it mean for a matrix A to have eigenvalue 0.

We know that.

$$Ax = \lambda x.$$

if $\lambda = 0$.

$$\Rightarrow Ax = 0 \cdot x$$

$$\Rightarrow \boxed{Ax = 0} \text{ --- (1)}$$

This suggests that (1) has non trivial solutions if and only if A is not invertible.

Note: (Explanation).

- A trivial sol. is $x=0$

- A nontrivial sol. mean that there is some $x \neq 0$ such that $Ax=0$.

If such nonzero sol exist, it means that 0 is an eigenvalue.

also recall eigen vectors cannot be zero.

$\therefore$ for eigen values to be 0.

$$Ax = 0$$

A^{-1} should not be possible

$$x = A^{-1}0 \Rightarrow x = 0$$

$\downarrow$

does not exist.

homogeneous system of equations.

$\therefore$ for 0 to be eigen value

Theorem no. 2: - A should be not invertible

If $v_1, \dots, v_r$ are eigen vectors that correspond to distinct eigen values $\lambda_1, \lambda_2, \dots, \lambda_r$ of an $n \times n$ matrix, then the set $\{v_1, \dots, v_r\}$ is linearly independent.



The scalar λ is an eigen value of A if it satisfies the characteristic eq.

EXERCISE 5.2

THE Characteristic Eqn.

Characteristic equation → a scalar eqn.
stores/stores special information about the eigenvalues of a square matrix A .

FINDING λ if A is given.

Example:

$$A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$$

$$A - \lambda I = 0$$

Now.

$$A - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix}$$

$$\det \begin{pmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$(2-\lambda)(-6-\lambda) - 9$$

$$-12 - 2\lambda + 6\lambda + \lambda^2 - 9$$

Note:

$$\lambda^2 + 4\lambda - 21 \rightarrow \det \text{ transformed}$$

$$\lambda^2 + 7\lambda - 3\lambda - 21$$

$$[(\lambda + 7)\lambda] - [3(\lambda + 7)]$$

$A - \lambda I$

↓ into

$$(\lambda + 7)(\lambda - 3) = 0 \quad \lambda^2 + 4\lambda - 21$$

$$\lambda = 3 \quad \lambda = 7 \quad \left. \begin{array}{l} \end{array} \right\} \text{Ans!}$$

Example:

Compute the det. of

$$\begin{bmatrix} 2 & 3 & 1 \\ 4 & 0 & -1 \\ 0 & 2 & 1 \end{bmatrix}$$

expansion from C_1 .

$$2 \det \begin{pmatrix} 0 & -1 \\ 2 & 1 \end{pmatrix} - 4 \det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$$

$$+ 0 \det \begin{pmatrix} 3 & 1 \\ 0 & -1 \end{pmatrix}$$



Note: A is invertible iff the equation $Ax=0$ has only the trivial solution. ^{day/date:}

$$2(0 - (-2)) - 4(3 - 2) + 0(-3 - 0) = 0$$

Theorem 3: Properties of determinants

Let A and B are $n \times n$ matrices.

a. A is invertible iff $\det A \neq 0$

b. $\det AB = (\det A)(\det B)$

* c. $\det A^T = \det A$

* d. If A is triangular the $\det$ of A is the product of the entries on the main diagonal of A .

e. $\rightarrow$ A row replacement op. does not change the $\det$.

$\rightarrow$ A row exchange/operation changes the (interchange) sign of the determinant.

$\rightarrow$ A row scaling also scales the $\det$ by the same scalar factor.

Theorem :-

The invertible Matrix theorem.

$\sim x \sim$
Previously we have proven how iff 0 is an eigenvalue then $Ax=0$ should have only non trivial sol while A is not invertible.

$\sim x \sim$
Now this theorem talks in opposite direction of the concept too $\rightarrow$ another view.

Let A be an $n \times n$ matrix. Then A is invertible iff

v. The number 0 is not an eigenvalue of A .

NOTE :-

$\det(A - \lambda I)$ is a polynomial in λ . It can be shown from that degree of this polynomial for a $n \times n$ A matrix is also n !

(iii) Say you have a resulting polynomial from an $\det(A - \lambda I)$ to be $(\lambda - 5)^2(\lambda - 3)(\lambda - 1)$

here:

eigen value 5 has a multiplicity of 2

as $\lambda - 5$ occurs 2 times

as the factor of characteristic polynomial

* $\therefore$ In general the Multiplicity of an eigen value λ is its multiplicity as a root of the characteristic eqn.

"SIMILARITY"

If A and B are two $n \times n$ matrices, then A is similar to B if there is an invertible matrix P such that $P^{-1}AP = B$ or equivalently $A = PBP^{-1}$

w.r.t Q for P^{-1}

$$A = Q^{-1}BQ$$

while here we can say that converting A into $P^{-1}AP$ is called as similarity transformation.

THEOREM 4.

If $n \times n$ matrices 'A' and 'B' are similar, then they have

- the same characteristic eqn.

and hence the same eigen values.
(with the same multiplicities).

① $\Rightarrow$ The matrices.

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \text{ and } \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

are not similar.

even if they have the same eigen value.

② Similarity is not the same as row equivalent (If A is row equivalent to B, then $B = EA$ for some invertible matrix E)

Row operations on a matrix usually change its eigen values.

Two matrices A and B are similar if they exist with an invertible matrix P such that.

$$B = P^{-1}AP$$

This means that they represent the same linear transformation just in different Bases.

If two matrices are similar; they have.

- (i) the same eigenvalues. ✓
- (ii) have the same algebraic and geometric multiplicities ✓
- (iii) And they are diagonalizable in the same way.

Geometric multiplicities refers to the number linear independent eigenvectors.

Algebraic multiplicity refers to how many times λ has appeared as the root of the characteristic polynomial of a matrix.

Note

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

↙
triangular.

$$\lambda = (2)^2$$

$$B = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \text{ does not have the}$$

↙
triangular.

$$\lambda = (2)^2$$

$$\boxed{A - \lambda I = 0}$$

Both A and B are not similar because if you solve
↙ ↘ 1 linearly independent eigen vector. we would have
2 linearly independent eigen vector

EXERCISE 5.3:

DIAGONALIZATION:-

The eigenvalues and eigenvectors information contained within a matrix A can be displayed in a useful factorization of the form $A = PDP^{-1}$.

where D : Diagonal Matrix

* This factorization also helps us to find A^k more quickly.

Example

$$D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

$$D^2 = \begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}$$

$$D^3 = \begin{bmatrix} 5^3 & 0 \\ 0 & 3^3 \end{bmatrix}$$

$$D^k = \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix} \text{ for } k \geq 1$$

Now

if $A = PDP^{-1}$
for some invertible P
and a diagonal D ,
then A^k is also easy
to compute

In general (check page 315 for e.g.)



$$A^k = P D^k P^{-1}$$

Note:

A square matrix A is said to be diagonalizable if A is similar to a diagonal Matrix i.e. ; $A = PDP^{-1}$

for some invertible matrix P and some diagonal matrix D .

THEOREM NO-5 DIAGONALIZATION THEOREM:-



The $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigen vectors.

Because;

$$A = PDP^{-1}$$

with D being a

Diagonal matrix.

if the columns of P are linearly independent eigen vectors of A .

Also D has the Diagonal entries equal to the eigen values of the A that corresponds respectively to the eigen vectors in P .

EXAMPLE 3.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

Step no. 1

finding eigen values

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{vmatrix} = 0$$

$$-\lambda^3 - 3\lambda^2 + 4 = 0$$

$$-(\lambda - 1)(\lambda + 2)^2 = 0$$

$$\lambda = 1$$

$$\lambda = -2$$

Step no. 2

Now find the linearly independent eigen vectors of A .

$$\lambda = 1$$

$$Ax_1 = 1x_1$$

$$(A - I)x_1 = 0$$

Solve

$$x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \text{ similarly.}$$

$$\text{for } \lambda = -2, x_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Now you can see that the set

$\Rightarrow \{X_1, X_2, X_3\}$ is a set of linearly independent Eigen vectors.

Step no. 3

Construct a P from the vectors in step no. 2.

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Important Computational technique.

We know

$$A = P D P^{-1} \quad \text{--- (1)}$$

$\Rightarrow AP = PD$. if we satisfy this we would eventually satisfy (1).

$$\therefore \overset{A}{\begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}} \overset{P}{\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}} \Rightarrow \begin{bmatrix} 1 & +2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

Similarly $PD \Rightarrow \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$ $\therefore$ we have successfully diagonalized our matrix.

for a basis in $\mathbb{R}^3$ (we should have 3 linearly independent eigenvectors). n -Basis ^{dim/dim} is a requirement
($\downarrow$ of eigenvectors)

THEOREM 6:-

An $n \times n$ matrix is diagonalizable if it has n distinct eigenvalues.

$n \times n$ matrix with n distinct eigenvalues is diagonalizable.

THEOREM NO. 7:-

Let A be an $n \times n$ matrix. Suppose A has distinct eigenvalues $\lambda_1, \lambda_2, \lambda_p, \dots$ (so it has p different eigenvalues)

$\hookrightarrow$ possible with some being repeated

(a) for each λ_k , the dimension of the eigen space is less than or equal to the algebraic multiplicity of λ_k .

$\Rightarrow$ In simple words

each eigenvalue may appear multiple times

P is invertible if the columns of A are linearly independent or if the columns of P are formed from the basis eigenvectors.

$\sim x \sim$

in the characteristic polynomial eqn that counts its algebraic multiplicity.

But, the no. of linearly independent eigenvectors $\rightarrow$ geometric multiplicity and dimension of eigenspace $\rightarrow$ for that eigenvalue can never be more than the algebraic multiplicity of λ_k .

geometric multiplicity of $\lambda_k \leq$ algebraic multiplicity of λ_k .

(b) A is diagonalizable iff the total no. of linearly independent eigenvectors is equal to n .



- (c) If A is diagonalizable and each eigenvalue λ_k (for) you have a basis B_k , for its eigen space, then putting all these basis of $\mathbb{R}^n$ made up of eigenvectors. Basis vectors together gives.

called as the eigenvalue of T if there is a non trivial solution of x

$T(x) = \lambda x$; such that an x is called as the eigen vector of the corresponding λ .

EXERCISE 504

DIAGONALIZATION & LINEAR TRANSFORMATION.

at this section:

eigen values and eigen vectors of linear transformation

$T: V \rightarrow V$. where V is any vector space.

DEFINITION.

Let V be a vector space, An eigenvector of a linear transformation $T: V \rightarrow V$ is a non zero vector x such that $T(x) = \lambda x$ where λ is a scalar quantity

Example:-

$$\{s_k\} = \left\{ \cos\left(\frac{k\pi}{2}\right) \right\}$$

$k \in$ all integers.

the double left shift linear transformation D is defined as.

$$D(\{x_k\}) = \{x_{k+2}\}.$$

Question:-

Show that $\{s_k\}$ is an eigen vector of transformation D and give its eigen value. ?

Note:

trigonometric formula:

we want to show that the signal s_k is the eigen vector of transformation of D .

$$D \{s_k\} = \lambda \{s_k\}.$$

we have also been given that

$$D \{x_k\} = \{x_{k+2}\}.$$



$$y_k \Rightarrow s_{k+2}$$

$$\Rightarrow \cos\left(\frac{(k+2)\pi}{2}\right)$$

$$\Rightarrow \cos\left(\frac{k\pi}{2} + \pi\right)$$

$$\Rightarrow -\cos\left(\frac{k\pi}{2}\right) = -s_k.$$

because.

$$\cos(\theta + \pi) = -\cos(\theta).$$

CONCEPT

AS MATRIX

WRITING TRANSFORMATION ↗

When we fix a basis, you can turn a linear transformation in a matrix.

That way, applying the transformations is the same as doing matrix multiplication.

How:-

- i) Pick a basis $B = \{b_1, b_2 \dots b_n\}$
- ii) compute ^{what} the transformation does to each basis vector.

$$T(b_1), T(b_2) \dots T(b_n).$$

Express each $T(b_i)$ as/in terms of the same Basis B .

stack those cols in a matrix, the matrix is the representation of Transformation T relative to B .

$$\begin{array}{ccc}
 x & \xrightarrow{\quad} & T(x) \\
 \downarrow & \text{multiplication} & \downarrow \\
 [x]_B & \xrightarrow{M=[T]_B} & [T(x)]_B
 \end{array}$$

Now using the concept defined earlier.

Let V be a vector ^{space} with
 ① Basis $B \Rightarrow \{b_1, \dots, b_n\}$.

Let x be a vector of V
 so it can be given as.

$$\begin{aligned}
 x &= r_1 b_1 + r_2 b_2 + \dots + r_n b_n \\
 \Rightarrow [x]_B &= \begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix}
 \end{aligned}$$

Now apply linear

② transformation T such that

$$\begin{aligned}
 T(x) &= r_1 T(b_1) + r_2 T(b_2) \\
 &+ \dots + r_n T(b_n).
 \end{aligned}$$

③ express each transformed Basis vector $T(b_i)$ and express it as the linear combination of the Basis B .

$[T(b_i)]_B \Rightarrow$ the coordinate vector of $T(b_i)$ in the Basis B

④ Stack them in the cols of the matrix.

$$M = \begin{bmatrix} [T(b_1)]_B & [T(b_2)]_B & \dots & [T(b_n)]_B \end{bmatrix}$$

The matrix M is the matrix representation of T in basis B .

$$[T]_B = M.$$

hence this is how we represent a transformation in terms of a matrix.

EXAMPLE:-

Suppose

$B = \{b_1, b_2\}$ is a basis for V .

Let $T: V \rightarrow V$

be a linear transformation with the property that

$$T(b_1) = 3b_1 - 2b_2$$

$$T(b_2) = 4b_1 + 7b_2$$



$$[T(b_1)]_B \Rightarrow \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$[T(b_2)]_B \Rightarrow \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

$$M \Rightarrow \begin{bmatrix} 3 & 4 \\ -2 & 7 \end{bmatrix} \text{ Ans!}$$

Example no. 3.

$$T \rightarrow \mathbb{P}^2 \rightarrow \mathbb{P}^2$$

$$T(a_0 + a_1 t + a_2 t^2) = a_1 + 2a_2 t$$

(a) find the B matrix of Transformation when the Basis are $\{1, t, t^2\}$.

let us use these basis to apply transformation at them.

Note that the transformations given to us as the derivative

$$T(b_1) \Rightarrow 0$$

$$T(b_2) \Rightarrow T(t) \Rightarrow 1$$

$$T(b_3) \Rightarrow T(t^2) \Rightarrow 2t$$

Now writing the transformations applied to Basis vector and express it as the coordinates in the Basis B .

$$[T(b_1)]_B \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$[T(b_2)]_B \Rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$[T(b_3)]_B \Rightarrow \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

$$[T]_B \Rightarrow M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T(p)_B = [B][p_B]$$

(b) to prove

$$[T(p)]_B = [T]_B [p_B]$$

$$[T(p)]_B \Rightarrow [a + 2a_1 t]$$

$$\rightarrow \begin{bmatrix} a_1 \\ 2a_1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

Note:

- There are infinite many eigenvectors but here we found 2 linearly independent eigenvectors (note from an example class, note for idea).

Diagonal Matrix Represent.

THEOREM 8.

Suppose $A = PDP^{-1}$ where

D is a diagonal $n \times n$ matrix. If P is the Basis for $\mathbb{R}^n$ formed from the columns of P , then

D is the B matrix of the Transformation $x \mapsto Ax$.

Example

Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$
by $T(x) = Ax$.

where $A = \begin{bmatrix} 7 & 2 \\ 4 & -1 \end{bmatrix}$

find a Basis B for $\mathbb{R}^2$
with the property the
 B -matrix of the transformation
is a diagonal matrix.

(from prev procedures we deduce that)

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}, D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

the columns of P being b_1 & b_2
i.e. the eigenvectors of A .

By theorem 8, D is the

B -matrix for T i.e

$$B = \{b_1, b_2\}$$

SIMILARITY OF MATRIX REPRESENTATION.

we know that
for two matrices to be
similar, let $\downarrow$
let A & C .

$$A = PCP^{-1}$$

we can also say here
as said before for
Diagonalization of matrix
that C

$$\text{By } A = PDP^{-1}$$

D is the
Basis
matrix

for the transformation
(e.g. $x \mapsto Ax$).



Note:

* If A and C are similar, it means they have the same linear transformation but in different Bases.

* Diagonalization is a special concept of similarity of Matrices.

"CHAPTER NO. 6"

Inner product.

Say.

Multiplication of two matrices u ($n \times 1$ matrix) & v ($1 \times n$ matrix) results in a scalar quantity.

→ this is called as the inner / dot product denoted as $u \cdot v$.

$$(c_1 u_1 + \dots + c_p u_p) \cdot w$$

$$= c_1 (u_1 \cdot w) + \dots + c_p (u_p \cdot w)$$

LENGTH OF A VECTOR.

The length / norm of v is the nonnegative scalar $\|v\|$ defined by

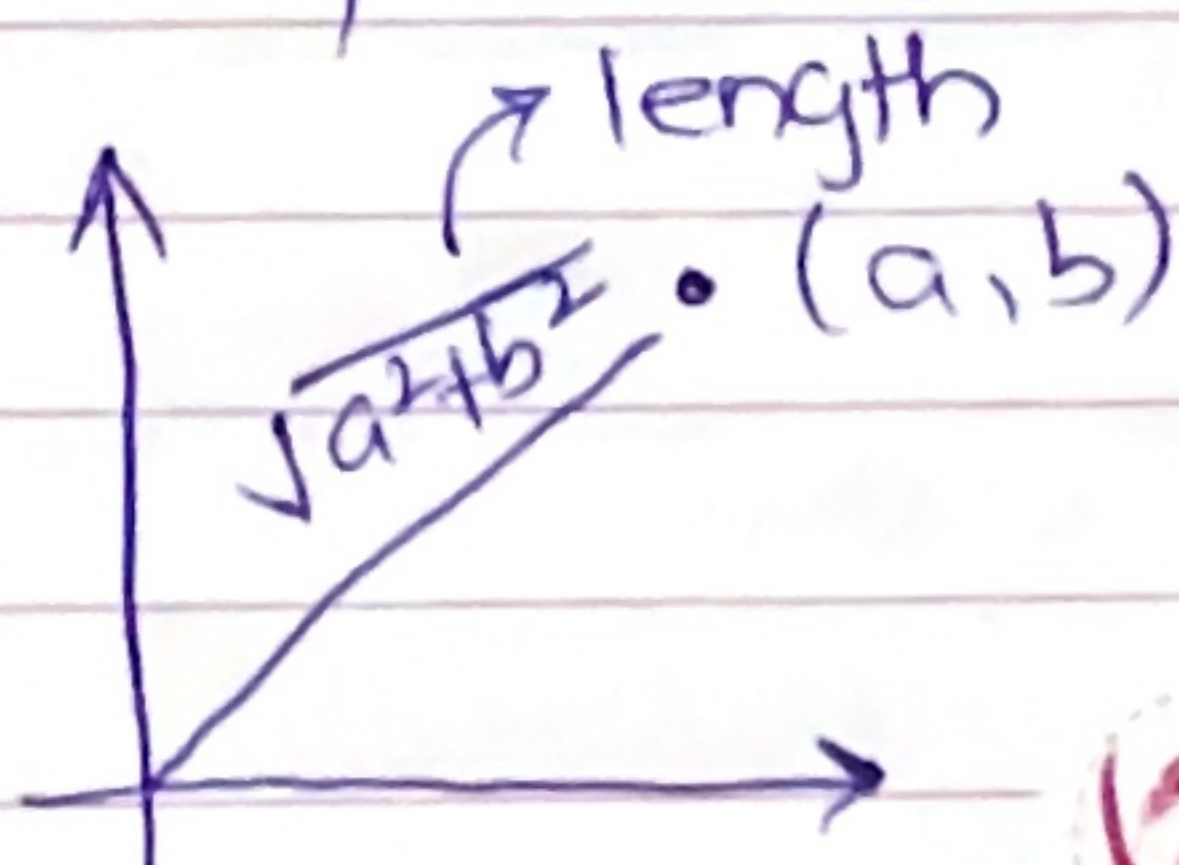
$$\|v\| = \sqrt{v \cdot v} \\ \Rightarrow \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

and

$$\Rightarrow \|v^2\| = v \cdot v$$

let v be $\begin{bmatrix} a \\ b \end{bmatrix}$ represented

in the plane as



THEOREM NO.1.

Let u, w & v be the vectors in $\mathbb{R}^n$ and let c be a scalar.

Then:

$$(a) \quad u \cdot v = v \cdot u$$

$$(b) \quad (u+v) \cdot w = u \cdot w + v \cdot w$$

$$(c) \quad (cu) \cdot v = c(u \cdot v) = u \cdot (cv)$$

$$(d) \quad u \cdot u \geq 0 \text{ and } u \cdot u = 0 \text{ iff } u = 0.$$

From theorem no.1 we can deduce.



* additionally note.

for any scalar 'c',
the length of $c\vec{v}$ is
 $|c|$ times the length
of $\vec{v}$. i.e

$$\|c\vec{v}\| = |c| \|\vec{v}\| \quad \text{--- (a)}$$

Similarly the length
of $(c\vec{v})^2 \Rightarrow$

$$\begin{aligned} \|c\vec{v}\|^2 &\Rightarrow c^2 \|\vec{v}\|^2 \\ &\Rightarrow c^2 \|\vec{v}\|^2 \quad \text{--- (b)} \end{aligned}$$

NORMALIZATION:-

Consider a vector $\vec{u}$.

Then the unit vector $\hat{u}$
can be given as

$$\hat{u} = \frac{\vec{u}}{\|\vec{u}\|}$$

also note.

$$\|\hat{u}\| = \frac{1}{\|\vec{u}\|} \times \|\vec{u}\| \Rightarrow 1 \quad \text{--- (c)}$$

EXAMPLE:-

Let $\vec{v} = (1, -2, 2, 0)$

find the unit vector ' $\vec{u}$ '
in the same direction
as $\vec{v}$.

$$\|\vec{v}\|^2 \Rightarrow (1)^2 + (-2)^2 + (2)^2 + (0)^2$$

$$\begin{aligned} \sqrt{\|\vec{v}\|^2} &\Rightarrow \sqrt{9} \\ &\Rightarrow 3. \end{aligned}$$

Now for the unit vector.

$$\vec{u} = \frac{1}{\|\vec{v}\|} \vec{v}$$

$$\Rightarrow \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \\ 0 \end{bmatrix}$$

to check that our unit
vector is correct solve.

$\|\vec{u}\|$ to be equal to 1.

→ P.T.O.

Example no. 3

Distance in $\mathbb{R}^n$

For 'u' and 'v' in $\mathbb{R}^n$,
the distance between
'u' and 'v' is written
as $\text{dist}(u, v)$, is the
length of the vector

$$u - v \text{ i.e.}$$

$$\text{dist}(u, v) = \|u - v\| \quad \text{--- ①}$$

Example no. 4.

Compute the distance
blw the vectors

$$u = (7, 1) \text{ and } v = (3, 2).$$

$$\|u - v\| = \begin{bmatrix} 7 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

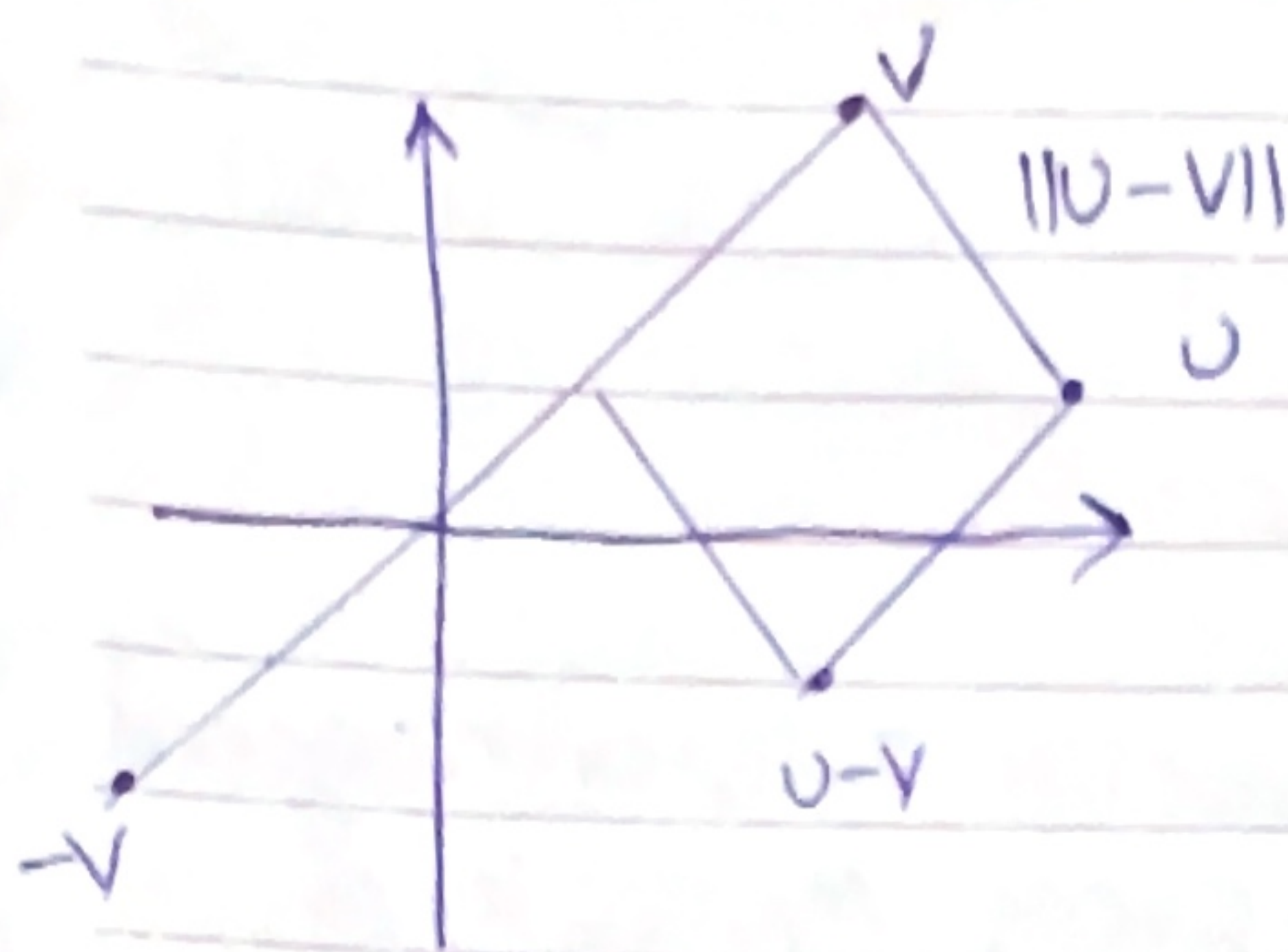
$$\Rightarrow \left\| \begin{bmatrix} 4 \\ -1 \end{bmatrix} \right\|$$

$$\Rightarrow \sqrt{(4)^2 + (-1)^2}$$

$$\Rightarrow \sqrt{16 + 1} \Rightarrow \sqrt{17} \text{ Ans}$$

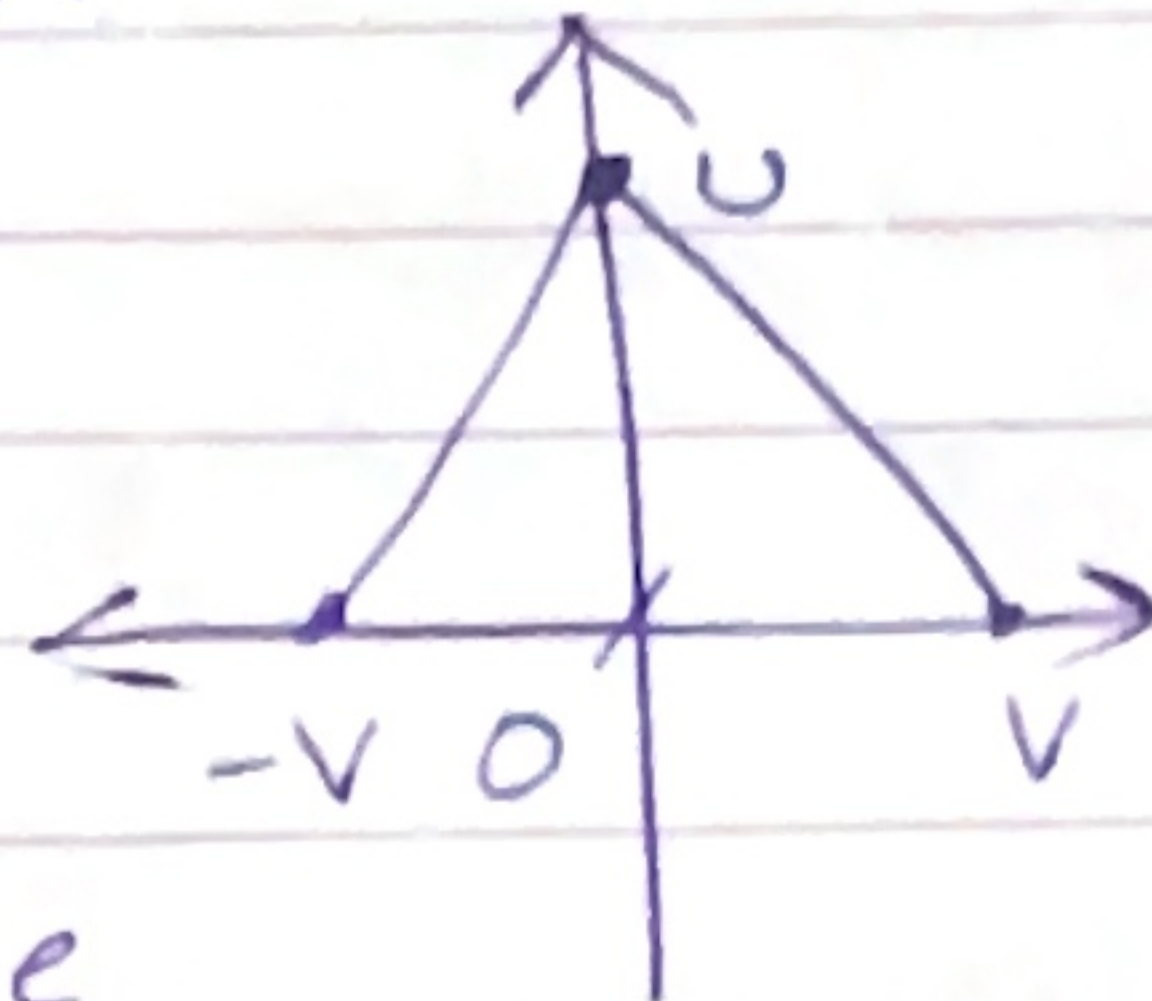
We can view them
geographically as ...





We then say:

Two vectors are perpendicular geometrically iff the distance from u to v is the same as from u to $-v$.



i.e.

$$\text{dist}(u, v) = \text{dist}(u, -v)$$

also Note, if you are given (imp eqns)

$$u = (u_1, u_2, u_3)$$

$$v = (v_1, v_2, v_3)$$

then

$$\text{dist}(u, v) = \|u - v\| \quad \text{--- (iii)}$$

$$\Rightarrow \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2} \quad \text{--- (ii)}$$

$$\Rightarrow \sqrt{(u - v) \cdot (u - v)} \quad \text{--- (i)}$$

$$\|u - v\| = \|u - (-v)\|$$

$$\Rightarrow \|u + v\|$$

$$\Rightarrow (u - v)(u - v) = (u + v)(u + v)$$

$$u^2 + v^2 - 2uv \Rightarrow u^2 + v^2 + 2uv$$

$$-u \cdot v = uv$$

possible only iff

$$u \cdot v = 0$$

↓

another condition of orthogonality.

ORTHOGONAL VECTORS!

How do we know if two vectors in $\mathbb{R}^n$ are orthogonal / perpendicular or not?

(it is easier in $\mathbb{R}^2$ but how in $\mathbb{R}^3$).

Two vectors u & v (orthogonal) to each other if $u \cdot v = 0$.